Fiscal Year:	FY 2013	**Task Last Updated:**	FY 06/04/2013
PI Name:	Hargens, Alan R. Ph.D.		
Project Title:	Fluid Distribution before, during and after Prolonged Space Flight		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92037-0863	**Congressional District:**	52
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Crew Health NNJ11ZSA002NA
Start Date:	07/11/2012	**End Date:**	01/10/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	Flight Definition		
Key Personnel Changes/Previous PI:	New Co-I: Brandon R Macias, PhD NSBRI Postdoctoral Fellow Department of Orthopaedic Surgery University of California, San Diego San Diego, CA 92103-8894 Email: b1macias@ucsd.edu		
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Grant/Contract No.:	NNX12AL66G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We will use state-of-the-art, non-invasive technologies to quantify upper-body compartmental volumes and pressures in crew members before, during and after prolonged space flight. Importantly, we will correlate these data with vision deficits that occur in order to establish pathophysiologic mechanisms that will serve as basis for future countermeasure development. After successful completion of our investigation, we will deliver a comprehensive database of microgravity-induced, head-ward volume and pressure changes (type and magnitude) and a prioritization of these changes as to their deleterious effects on vision in crewmembers during and after prolonged space flight. We are proposing a well-documented and validated battery of non-invasive or minimally-invasive, image-based tests developed to identify and quantify microgravity-induced, head-ward volume and pressure changes. We hypothesize that prolonged microgravity-induced, head-ward volume and pressure shifts are responsible for elevating intracranial pressure (ICP) and producing deficits in crewmembers' vision. Our project directly addresses Critical Path Roadmap Risks and Questions regarding "Risk of Microgravity-Induced Visual Alterations and Intracranial Pressure", specifically IRP Gap CV7: How are fluids redistributed in-flight? and IRP Gap VIIP1: What is the etiology of visual acuity and ocular structural and functional changes seen in-flight and post-flight? Our first specific aim is to study periocular fluid volumes, intraocular pressure (IOP), upper-body compartment volumes before, during and after prolonged microgravity exposure. The second specific aim is to measure jugular vein dimensions and blood flow using ultrasound before, during and after prolonged microgravity exposure. The third specific aim is to quantify ventricular and cerebrospinal volumes using ultrasound before, during and after prolonged microgravity exposure. A fourth specific aim is to perform retinal imaging to observe retinal venous distension in space. Tests of ocular structure will include optic nerve head tomography, nerve fiber layer thickness, axial length, and orbital retrolaminar subarachnoidal space. Tests of ocular function will include visual acuity, total retinal blood flow, and capillary blood flow in the optic nerve head and macula. Finally, changes in ICP, IOP, and ocular structures and functions will be investigated while applying a purely-mechanical countermeasure of low-level lower body negative pressure or thigh cuffs to counteract the head-ward fluid shift in space. To our knowledge, this study will be the first to provide detailed and non-invasive measures of compartmental volume and pressure changes in the upper body induced by prolonged microgravity and to correlate these specific changes with decrements in vision for crewmembers. The proposed techniques represent the best available, state-of-the-art tools to quantify and document features that are clinically suspected as vision deficit generators. By correlating volume and pressure changes with vision problems, we expect to identify factors that will later motivate targeted development of effective physiologic countermeasures such low-level lower body negative pressure exposure or thigh cuffs in space. This project has the potential to prevent loss of vision in crewmembers exposed to prolonged space flight and upon return to Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A focus on fluid distribution is a very high research priority that addresses most physiologic systems of the body and is particularly relevant for humans because of the large gravitational pressure gradients involved. Studies will help determine the basic mechanisms, adaptations, and clinical significance of changes in regional vascular and tissue compartment pressures (Starling pressures and lymph function) for patients with elevated intracranial pressure (ICP) and glaucoma. These changes may be especially important in the brain and around the optic nerve and other ocular structures, where chronic and sustained increases in intracranial, intraocular, and retinal pressures may induce clinically-important structural and functional abnormalities that alter visual acuity. An example is applying a purely-mechanical countermeasure of low-level LBNP to counteract the head-ward fluid shift in patients with high ICP.
Task Progress:	EDITOR'S NOTE (4/24/2013): PROJECT CONTINUES AS "Fluid Distribution before, during and after Prolonged Space Flight--NNX13AJ12G" WITH THE SAME PRINCIPAL INVESTIGATOR; PERIOD OF PERFORMANCE IS 4/5/2013-9/30/2018. SEE THAT PROJECT FOR SUBSEQUENT REPORTING.
Bibliography Type:	Description: (Last Updated: 08/25/2025)